

TechCorp Disaster Recovery Plan

Incident Classification, RTO/RPO Targets & Recovery Procedures

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TechCorp Internal

1. Incident Classification

Class	Scope	Examples	Declares DR
Class 1	Single service failure	Pod crash, API timeout	No — SRE handles
Class 2	Multi-service or regional impact	AZ outage, DB failover needed	Optional
Class 3	Full production unavailability	Region failure, major data loss	Yes — immediate
Class 4	Security breach	Data exfiltration, ransomware	Yes + Legal

2. Recovery Time Objectives (RTO)

Core API (production)	RTO: 30 minutes
Web Application	RTO: 30 minutes
Database (primary)	RTO: 15 minutes (Aurora auto-failover)
RAG Pipeline	RTO: 1 hour
Background Jobs	RTO: 2 hours
Analytics / Reporting	RTO: 4 hours
Internal Tools	RTO: 8 hours

3. Recovery Point Objectives (RPO)

Transaction Database	RPO: 5 minutes (Aurora PITR + WAL streaming)
Document Store	RPO: 15 minutes (S3 versioning + replication)
Vector Database (Qdrant)	RPO: 1 hour (hourly snapshots to S3)
Audit Logs	RPO: 0 minutes (CloudWatch Logs — real-time)
Embedding Cache	RPO: 24 hours (rebuildable from source)
Analytics Data	RPO: 1 hour (Kinesis Firehose delivery)

4. Backup Strategy

- Aurora: automated backups enabled, 35-day retention, PITR to 5-minute granularity
- S3 documents: versioning + cross-region replication to eu-west-1 with 30-day lifecycle
- Qdrant vector DB: hourly collection snapshots uploaded to S3 with 7-day retention
- Redis cache: Redis AOF persistence + 6-hourly RDB snapshot to S3
- Infrastructure as Code: all Terraform state in S3 with versioning — no manual infra
- Application code: GitHub — all commits immutable, branch protection on main
- Secrets: AWS Secrets Manager with cross-region replication enabled

5. Database Recovery

Step 1: Assess damage – read Aurora cluster events in AWS Console Step 2: If primary AZ down → Aurora auto-promotes read replica (ETA: 60-120s) Step 3: If cluster unrecoverable → restore from latest automated snapshot Step 4: Verify data integrity: run db-health-check.sh on restored cluster Step 5: Update connection strings in Secrets Manager to point to new endpoint Step 6: Restart application services to pick up new connection Step 7: Monitor for replication lag and connection pool errors

6. Infrastructure Recovery

Step 1: Declare DR incident – notify VP Eng, SRE Lead, Security Step 2: Activate DR runbook in Notion: [link to DR runbook] Step 3: If us-east-1 unhealthy → update Route 53 failover records to eu-west-1 Step 4: Scale up eu-west-1 ECS services: ./scripts/scale-dr-region.sh eu-west-1 Step 5: Verify health checks pass in DR region Step 6: Communicate status on status.techcorp.io Step 7: Once primary region stable → plan fail-back window (min 24h after recovery)

7. Communication Plan

T+0	SRE declares incident; opens war room bridge; pings #incident-response
T+15m	Initial customer communication posted on status.techcorp.io
T+30m	Account executives notify Enterprise customers directly (email + call)
T+1h	Update on status page — root cause hypothesis + ETA for resolution
T+2h	Exec briefing — VP Eng + CTO updated on status and business impact
Post	Resolution notice on status page; post-mortem published within 5 days

8. DR Testing Schedule

Test Type	Frequency	Owner	Success Criteria
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Backup Restore Drill	Monthly	SRE	Restore within RTO, data matches RPO
Failover Test (Aurora)	Quarterly	DBA + SRE	Auto-failover < 120s, no data loss
Full Region Failover	Semi-annual	SRE + Eng	DR region live within 30 min
Runbook Review	Quarterly	SRE Lead	All steps accurate + tested
Tabletop Exercise	Annual	All teams	Org ready for Class 3/4 incident